

NEPHROTIC = PEEL THE KIDNEY
 $P =$ (proteinuria icon)
 $E =$ (edema icon)
 $A =$ (albuminuria icon)
 $L =$ (lipiduria icon)
>3.5 g/day

ALBUMIN LOST

THROMBOSIS - $\downarrow$ ATIII / Prot C+5
DYSLIPIDEMIA $\uparrow$ CVD
INFECTION - Ig LOSS
ALTERED DRUG TBG / VIT D
THROMBOSIS - $\downarrow$ ATIII / Prot C+5
ANTICOAG (EARE-BY-CASE)
STATIN - CVD
ACEI/ARB
SGLT2i
GENERAL TREATMENT PHARMACY CART

PODOCYTE ELEMENTARY
70-90% PEDIATRIC NEPHROTIC
10-15% ADULT
NIL LESION - LM NORMAL
EM: FOOT PROCESS EFFACEMENT
PREDNISONE - 1st LINE
RELAPSE 70-75%
ANTI-NEPHRIN / B-CELL ROLE
FOOT PROCESS EFFACEMENT

COVERED WALL/IAKEY

ESKD #1 PRIMARY GN US - 50% IN 10 YRS
SHALLOW BIOPSY = MISS DX
FSGS BASEMENT - CORTICOMEDULLARY JCT
APOL1 RISK - AFRICAN ANCESTRY
PRIMARY $\rightarrow$ STEROIDS + CNI
SECONDARY GENETIC $\rightarrow$ NO STEROIDS - TREAT CAUSE
STERIODS POOR RESPONSE 20-45%
CNI (Cyclosporine)
RTX / MMF - UN-CERTAIN

NOS - CLASSIC
TIP - BEST PROGNOSIS / STEROID-RESPONSIVE + TUBULAR POLER
CELLULAR
PERIHILAR - SECONDARY HYPERFILTRATION STEAM JET
COLLAPSING - HIV / APOL1 / WORST

1. PEAL Mnemonic

WHAT YOU SEE

Chalkboard: P=E=A=L, upper left

WHAT IT ENCODES

PEAL summarizes the nephrotic tetrad: Proteinuria (>3.5 g/day), Edema, hypoAlbuminemia, hyperLipidemia (with lipiduria). The unifying defect is podocyte injury and loss of the glomerular charge/size barrier.

BOARD TRAP

Massive proteinuria is the driver — edema, low albumin, and hyperlipidemia are all downstream consequences, not independent diseases.

2. Proteinuria >3.5 g/day

WHAT YOU SEE

Chalkboard threshold

WHAT IT ENCODES

Nephrotic-range proteinuria defined as >3.5 g/24h. Quantify with 24h collection or spot urine protein:creatinine ratio. Magnitude correlates with progression risk and is itself nephrotoxic to tubules.

BOARD TRAP

Use the protein:creatinine ratio for convenience, but >3.5 g/day is the nephrotic threshold — sub-nephrotic with hematuria is nephritic instead.

3. Thrombosis (↑ ATIII / Protein Loss)

WHAT YOU SEE

Red clot figure, complications

WHAT IT ENCODES

Urinary loss of antithrombin III and altered clotting factors create hypercoagulability; renal vein thrombosis is most classic with membranous nephropathy, plus DVT/PE. Prophylactic anticoagulation when albumin is very low.

BOARD TRAP

Sudden flank pain + hematuria + worsening proteinuria in a membranous patient = renal vein thrombosis — anticoagulate.

4. Dyslipidemia / CVD

WHAT YOU SEE

Complications icon

WHAT IT ENCODES

Low oncotic pressure stimulates hepatic lipoprotein synthesis, raising LDL/triglycerides and accelerating atherosclerosis. Statins for persistent nephrotic dyslipidemia.

BOARD TRAP

Hyperlipidemia is reactive to hypoalbuminemia — it remits with disease control, but treat if persistent given CVD risk.

5. Infection / Altered Drug Binding

WHAT YOU SEE

Complications icons

WHAT IT ENCODES

Loss of immunoglobulins and complement factors increases infection risk (encapsulated organisms, SBP in children). Loss of binding proteins alters drug levels and thyroxine/vitamin D status.

BOARD TRAP

Nephrotic immunodeficiency comes from urinary IgG loss — vaccinate against pneumococcus; abdominal pain in a nephrotic child may be peritonitis.

6. General Treatment Pharmacy Cart

WHAT YOU SEE

Cart: statin, anticoag, RAAS, diuretic

WHAT IT ENCODES

Backbone nephrotic management: ACE inhibitor/ARB (lower proteinuria + BP), loop diuretic + salt restriction (edema), statin (dyslipidemia), anticoagulation (high thrombotic risk), plus cause-specific immunosuppression.

BOARD TRAP

RAAS blockade is universal antiproteinuric therapy, but immunosuppression must be tailored — steroids help MCD/primary FSGS, harm diabetic/amyloid nephropathy.

7. Minimal Change Disease

WHAT YOU SEE

Podocyte Elementary classroom

WHAT IT ENCODES

MCD: 70–90% of pediatric and 10–15% of adult nephrotic syndrome. LM normal, IF negative, EM shows diffuse foot-process effacement. Highly steroid-responsive ('prednisone → remission'). Secondary triggers: NSAIDs, Hodgkin lymphoma.

BOARD TRAP

In a child, treat empirically with steroids without biopsy; MCD is selective albuminuria — non-selective protein loss suggests another lesion.

8. Foot Process Effacement

WHAT YOU SEE

EM diagram in classroom

WHAT IT ENCODES

Diffuse podocyte foot-process effacement on EM is the unifying ultrastructural lesion of MCD (and seen in FSGS). Represents podocyte injury and loss of slit-diaphragm integrity, the cause of selective protein leak.

BOARD TRAP

Effacement is non-specific (MCD and FSGS share it) — the distinguishing feature is segmental sclerosis present in FSGS, absent in MCD.

9. Steroid Relapse / Anti-Nephrin Antibody

WHAT YOU SEE

Right panel, relapse note

WHAT IT ENCODES

MCD frequently relapses; frequent relapsers/steroid-dependent cases use calcineurin inhibitors, cyclophosphamide, or rituximab. Newly recognized anti-nephrin antibodies implicated in some MCD/podocytopathy (B-cell role supports rituximab).

BOARD TRAP

Steroid resistance in presumed MCD should prompt re-biopsy for FSGS — true MCD is steroid-sensitive even if relapsing.

10. FSGS Variants Lineup

WHAT YOU SEE

Bottom wall: NOS/Tip/Cellular/Perihilar/Collapsing

WHAT IT ENCODES

FSGS histologic variants: NOS (classic, most common), Tip (best prognosis, steroid-responsive), Cellular, Perihilar (hyperfiltration/secondary adaptive), and Collapsing (worst prognosis; HIV/APOL1). Variant predicts response and outcome.

BOARD TRAP

TIP variant = best prognosis; COLLAPSING = worst (HIV/APOL1) — perihilar suggests secondary/adaptive FSGS treated by removing the driver, not steroids.

11. APOL1 Risk / African Ancestry

WHAT YOU SEE

Stairway sign, APOL1 risk

WHAT IT ENCODES

Two APOL1 risk alleles (high in African ancestry) markedly increase risk of primary FSGS, collapsing FSGS, HIVAN, and hypertension-attributed CKD. Explains epidemiology of FSGS as #1 primary nephrotic cause in adults.

BOARD TRAP

APOL1 high-risk genotype underlies several podocytopathies — it's a genetic susceptibility, not itself the histologic diagnosis.

12. Primary vs Secondary FSGS

WHAT YOU SEE

Signpost: primary steroids vs secondary treat cause

WHAT IT ENCODES

Primary FSGS (diffuse effacement, abrupt nephrotic syndrome) responds to steroids/immunosuppression. Secondary/adaptive FSGS (obesity, reflux, reduced nephron mass, sickle cell — segmental effacement, slower onset) is treated with RAAS blockade and removing the driver.

BOARD TRAP

Don't immunosuppress secondary FSGS — high-dose steroids only help PRIMARY FSGS; adaptive FSGS needs RAAS blockade and cause control.

13. Effacement: Diffuse vs Segmental

WHAT YOU SEE

Basement EM placards

WHAT IT ENCODES

EM degree of foot-process effacement distinguishes primary (diffuse, widespread) from secondary (segmental, partial) FSGS — a key board discriminator guiding whether to immunosuppress.

BOARD TRAP

Diffuse effacement → primary FSGS (steroids); segmental/partial effacement → secondary/adaptive FSGS (RAAS + treat cause).

14. ESKD #1 — Primary GN ~50% in 10 yrs

WHAT YOU SEE

Red ESKD banner, bottom right

WHAT IT ENCODES

Untreated/steroid-resistant primary FSGS progresses to ESRD in roughly 50% within 10 years — the leading primary glomerular cause of ESRD. Collapsing variant progresses fastest. Can recur in transplant allografts (circulating permeability factor).

BOARD TRAP

Primary FSGS notoriously RECURS post-transplant (sometimes within hours, suPAR/permeability factor) — counsel and monitor early proteinuria in the graft.